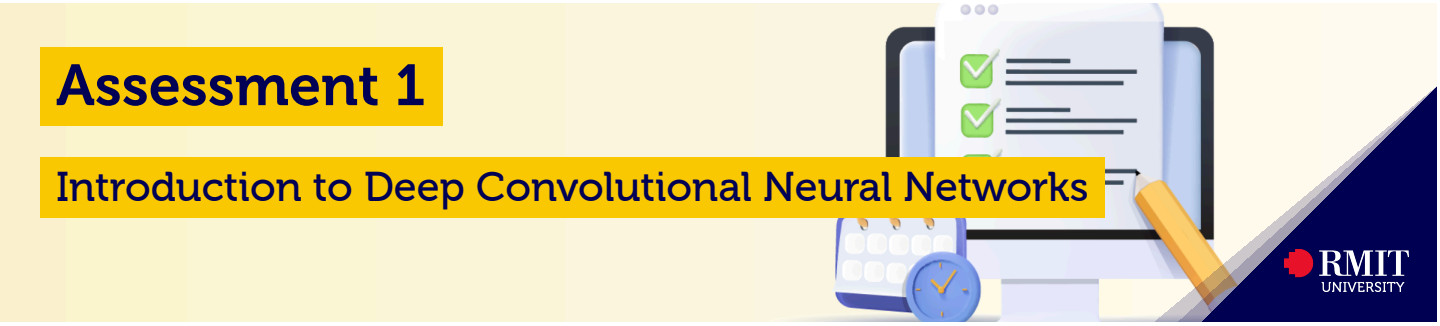
Assignment 1: Instructions and Notebook Submission



Course Code: COSC2792/COSC2779

Weighting: This assessment is worth 30% of your final mark for this Course

Length: No explicit page limit.

Task type: Individual

Due date and time: Check out the Syllabus page

Feedback: Feedback will be provided via Canvas

Click or tap on the relevant sections to learn more

1. Course Learning Outcomes Assessed

This assessment supports the following learning outcomes:

- CLO1: Discuss and critically analyse a variety of neural network architectures; Evaluate and compare approaches and algorithms based on the nature of the problem/task being addressed.
- CLO2: Synthesise suitable solutions to address particular machine learning problems based on analysis of the problem and characteristics of the data involved.
- CLO3: Communicate effectively with a variety of audiences through a range of modes and media, in particular to: interpret abstract theoretical propositions, choose methodologies, justify conclusions and defend professional decisions to both IT and non-IT personnel via technical reports of professional standard and technical presentations.
- CLO4: Develop skills for further self-directed learning in the general context of neural networks and machine learning; Research, Discuss, and use new and novel algorithms for solving problems; Adapt experience and knowledge to and from other computer science contexts such as artificial intelligence, machine learning, and software design.

2. Overview of Assessment

In this assignment, you will explore a real dataset to practice the typical deep learning process. The assignment is designed to help you become more confident in applying deep learning approaches. In this assignment, you will:

- Develop a deep learning system to solve a real-world problem.
- Analyse the output of the algorithm(s).
- Research how to extend the DL techniques that are taught in class.
- Provide an ultimate judgement of the final trained model that you would use in a real-world setting.

To complete this assignment, you will require skills and knowledge from lecture and lab material for Weeks 1 to 6 (inclusive). You may find that you will be unable to complete some of the activities until you have completed the relevant lab work. However, you will be able to commence work on some sections. Thus, do the work you can initially, and continue to build in new features as you learn the relevant skills. *A deep learning model cannot be developed within a day or two. Therefore, start early.*

3. Assessment Task

3.1 The situation

You've just joined RedKite Aerospace Australia, a young drone-manufacturing startup. Most of your colleagues are mechanical and aerospace engineers who can build an airframe that flies further, lifts more, and lands more precisely than anything else on the market. You are the company's first and, for now, the only machine learning engineer.

RedKite is chasing the contract that everyone in the building is talking about. A regional emergency-services provider has issued a request for proposal (RFP) for a drone capable of delivering time-critical medical supplies (i.e., blood products) to sites that ground vehicles can't reach quickly enough. RedKite's aircraft already meets every hardware requirement in the RFP. Range, payload, and endurance are all solved. The executives believe winning this contract puts RedKite on a path to real financial stability. Losing it means an uncertain year ahead.

There is one capability RedKite's product does not yet have, and the RFP makes it mandatory. When the drone arrives at a delivery site, it must follow the person receiving the package's hand-signal instructions (similar to marshalling-style gestures used to guide aircraft). The receiver waves the drone left, tells it to hold, signals when the landing zone is clear, or waves it off entirely if the site suddenly isn't safe to approach. No radio link, no app, just a person and their hands.

There's a demo in three weeks. On the day, a member of the client's staff will stand in the landing zone and signal to the drone. If RedKite can show the aircraft correctly reading that person's gestures, the contract is winnable. If it can't, the opportunity is gone.

Your job is to build this capability. When the team heard this, everyone volunteered. Over one afternoon, your colleagues stood in front of a camera and performed the gestures so you'd have something to train on. It is not a large dataset, and they are not professional signallers, but it is what you have, and the demo date isn't moving.

3.2 Part A — Build the classifier

The executives know three weeks and a small dataset is a hard hand. What they want to see is **sound engineering judgement** and a straight answer about **how far the result can be trusted**.

The onboard perception stack already includes a segmenter (provided; you do not build it) that watches the receiver and cuts the video into discrete instances — one complete gesture performance each. Each instance reaches your model as exactly 5 frames.

Your job is the classifier: given one pre-segmented instance (its 5 frames), predict which of the 13 gestures it is.

The 13 gesture classes are: AllClear, HaveCommand, Hover, Land, LandingDirection, MoveAhead, MoveDownward, MoveToLeft, MoveToRight, MoveUpward, NotClear, SlowDown, WaveOff

You do not need to detect the person, segment the video, or handle a live stream. One instance in, one class out.

Training Dataset: Access via **AWS** (instructions provided in lecture - Week 3)

Constraints

The onboard computer is already running navigation, mapping, and comms. The hardware team can spare enough compute for one modest 2-D CNN image classifier and nothing more.

- 2-D convolutional backbones only. No recurrent models (RNN/LSTM/GRU), no 3-D convolutions, and no temporal-attention / video-transformer architectures.
- The model must run inference on one instance at a time on an embedded board. Keep it small enough that real-time operation on constrained hardware is plausible, and argue that it is.
- You may use ImageNet-pretrained backbones and standard augmentation. You may not pretrain on external gesture or action-recognition datasets.

Everything else, including how you turn five frames and a 2-D CNN into a working gesture classifier, and how you convince yourself it will hold up on demo day, is for you to decide.

3.3 Part B — The recommendation

Assume your demo succeeded. The executives have congratulated you, signed off your probation, and given you a raise. The client is delighted and has asked RedKite to deliver the first operational units within two months.

Because the demo worked, **the client believes no further development or testing is required**: they want the exact gesture-recognition model you built in three weeks, on limited data, shipped into a safety-critical deployment as-is.

Your executives, who are extremely busy and unwilling to read long text, have asked for your honest opinion on this plan before they reply to the client.

Write an executive summary, in point form, giving your recommendation on shipping the model as-is: the key challenges and risks, and what (if anything) you would want in place first.

Requirements:

- Keep it to less than a single page, 5-10 bullets maximum. Anything longer will be skimmed and your advice lost.
- Ground every point in evidence from your Part A work. Unsupported generalities score poorly.
- This is not a rerun of your Part A limitations analysis. Part A is your technical self-assessment; Part B is a decision recommendation for a non-technical executive. We are assessing whether you can translate technical understanding into a clear, honest business recommendation.
- Consider the problem broadly (e.g., technical, operational, and safety/accountability dimensions) but lead with what matters most to the decision.

3.4 Assessment Guidelines

What your notebook must document (Markdown)

You are building a deep-learning system, and every element of it must be justified using data analysis, performance analysis, and/or relevant literature (not merely asserted). Document this clearly in Markdown, at the point in the notebook where each decision is made:

- Methodology: For each element of your system, explain why it is included and the evidence supporting it.
- Evaluation framework: Explain how you selected your performance measure(s) and how you determined your data splits. Your reported performance should reflect what you can actually expect at deployment.
- Results & ultimate judgement: Analyse your model's behaviour with appropriate techniques, assess how well it performs the task in the real world, and discuss its limitations honestly.

Technical requirements

- You must use **PyTorch**, and the notebook must run on the **AWS platform** introduced in class. A notebook that does not run or is not using PyTorch scores zero.
- Your submission must execute cleanly via Restart & Run All, and must set a fixed random seed so your results are reproducible.

- All decision-supporting figures must be embedded in the notebook: learning curves, distributions, confusion matrices, and anything else you rely on to make or defend a choice. Do not depend on TensorBoard or Weights & Biases; the marker may not be able to open them.
- Neural-network techniques only. No SVM, random forests, or other classical ML.
- A clearly identifiable, self-built segment. You may use pretrained networks, but there must be network segment(s) that you designed and trained, and your code must show how that segment was developed and tuned.
- Higher grades (CR/DI/HD) require a network that uses all the information available in an instance (all five frames), not a single frame.

▼ 4. Timeline

To help you prepare for the assessment, please consider the following activities:

Timeline	Activities
Week 03	• Assignment specs will be released, and the introduction will be done in class. • You can read the specs and start data analysis.
Week 04	• Explore data and read literature that attempts to solve problems similar to the one in Assignment 1. • Establish performance measures, and explore trivial benchmarks. • Work on the overall experimental framework design.
Week 05	• Start coding, data pipeline, and baseline; • Model improvement by reading the outputs from the model.
Week 06	• Final model, analysis, and composing the final submission. • Check your solution against the methodology in Week 6 and adjust accordingly.

▼ 5. Assessment submission and feedback

Submit your work as a Jupyter notebook via the upload link on this Canvas page. The notebook must:

- contain both Part A and Part B;

- have all outputs (figures, tables, printed results) intact and visible; and
- run top-to-bottom after a kernel reset on the AWS platform without errors, if a marker chooses to re-run it.

You must also submit a PDF export of the same notebook via **Assignment 1: PDF Notebook upload** [Turnitin], which is used for similarity checking. The PDF must exactly match the submitted notebook. Any discrepancy between the two will result in a mark of zero for the assignment.

Assignments received late and without prior extension approval or special consideration will be penalised by a deduction of 10% of the total score possible per calendar day late for that assessment.

Feedback will be provided via Canvas.

▼ 6. Assessment Criteria

Your submission is marked against the three components below. The Criteria column lists the elements each component is judged on; the full grade-level descriptors (HD+, HD, DI, CR, PA, PA−, NN) for every component are given in the accompanying marking rubric at the bottom of this section.

This assessment is pitched at a high Bloom's level: **rather than following prescribed steps, you are expected to make your own reasoned judgments about how to solve this problem**. For this reason, each criterion below is stated at a high level. The detailed sub-elements the markers look for sit beneath each one, but they are deliberately not revealed at this stage. Doing so would give away the very design decisions you are being assessed on identifying. They will be shared with you alongside your feedback after marking.

Component	Criteria	Weighting
Approach (Part A)	1) Data exploration leading to a well-informed approach. 2) Identifying an adequate evaluation framework tailored to the problem. 3) Well-justified network architecture and objective. 4) Hyper-parameter selection strategy. 5) Approach satisfies all the stated requirements and restrictions.	60%

Component	Criteria	Weighting
Ultimate Judgment, Performance & Analysis (Part A)	1) Analysis of the model and its outputs using suitable methods. 2) A clear ultimate judgment is made. 3) The rationale behind the ultimate model is clear and considers all relevant aspects. 4) Limitations of the model are identified. 5) Realistic assessment of the model's chance of success at deployment.	25%
Part B — Executive Recommendation	1) A clear, actionable recommendation on the client's plan to ship the model as-is. 2) Every point is grounded in evidence from your own Part A analysis. 3) Appropriate breadth of risk: technical, operational, and safety/accountability. 4) Concise, point-form, and pitched for a busy, non-technical executive audience.	15%

Detailed Rubric: [rubric-assignment1_2026.pdf]

7. Academic Integrity

[Study Support Hub]: Academic skills advisors and librarians will help you with researching your topic, finding information, improving your academic writing skills, academic integrity, effective learning strategies, statistics, and more!

[Academic Integrity/plagiarism]: You can achieve academic integrity by honestly submitting work that is your own. Presenting work that fails to acknowledge other people's work within yours can compromise academic integrity.

[Special consideration]: If unexpected circumstances beyond your control have affected your ability to complete an assessment, you may be eligible to apply for special consideration.

For more information, please consult the course guide or contact your course coordinator.